

start : program

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program : program unit

| unit

;

unit : variable_decl

| func_definition

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func_definition : type_specifier ID LPAREN param_list RPAREN compound_statement

| type_specifier ID LPAREN RPAREN compound_statement

;

param_list : param_list COMMA type_specifier ID

| param_list COMMA type_specifier

| type_specifier ID

| type_specifier

;

compound_statement : LCURL statements RCURL

| LCURL RCURL

;

variable_decl : type_specifier declaration_list SEMICOLON

;

type_specifier : INT

| FLOAT

| VOID

| CHAR

;

declaration_list : declaration_list COMMA ID

| declaration_list COMMA ID LTHIRD CONST_INT RTHIRD

| ID

| ID LTHIRD CONST_INT RTHIRD

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statements : statement

| statements statement

;

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statement : variable_decl
           | expression_statement
           | compound_statement
           | FOR LPAREN expression_statement expression_statement expression RPAREN
             statement
           | IF LPAREN expression RPAREN statement
           | IF LPAREN expression RPAREN statement ELSE statement
           | WHILE LPAREN expression RPAREN statement
           | PRINTLN LPAREN ID RPAREN SEMICOLON
           | RETURN expression SEMICOLON
;

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expression_statement : SEMICOLON
                    | expression SEMICOLON
;

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variable : ID
         | ID LTHIRD expression RTHIRD
;

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expression : logic_expression
           | variable ASSIGNOP logic_expression
;

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logic_expression : rel_expression
                 | rel_expression LOGICOP rel_expression
;

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rel_expression : simple_expression
               | simple_expression RELOP simple_expression
;

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simple_expression : term
                 | simple_expression ADDOP term
;

```

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term : unary_expression
     | term MULOP unary_expression
;

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unary_expression : ADDOP unary_expression
                 | NOT unary_expression
                 | factor_info
;

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factor_info : factor
;

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factor : variable

- | ID LPAREN argument_list RPAREN
- | LPAREN expression RPAREN
- | CONST_INT
- | CONST_FLOAT
- | variable INCOP
- | variable DECOP

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argument_list : arguments

|
;

arguments : arguments COMMA logic_expression

| logic_expression
;